

# Adrien Rozario

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## EDUCATION

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**B.S. Information Science, Minor in Data Science**, *University of Maryland, College Park* Expected December 2026

## EXPERIENCE

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**Service Desk Technician**, *University of Maryland Engineering IT Help Desk* 2024-present

- Analyzed and resolved 900+ technical support tickets, identifying recurring system issues and contributing to workflow improvements within a multi-platform computing environment.
- Diagnosed hardware, software, account access, and network issues across Windows, macOS, and Ubuntu systems, supporting students, faculty, staff, and research infrastructure.
- Collaborated with cross-functional IT teams to maintain security, access control, and data protection standards while assisting with technical troubleshooting and operational support.
- Maintained detailed technical documentation and service records to support knowledge transfer, reproducibility, operational efficiency, and long-term system reliability.
- Communicated technical issues and solutions clearly to non-technical users in a fast-paced customer support environment.

**Service Desk Assistant**, *University of Maryland Engineering IT Help Desk* 2023-2024

- Supported service desk operations by assisting with hardware preparation, inventory management, workspace organization, and technical logistics within a fast-paced university IT environment.
- Prepared computing equipment for recycling and redistribution through collaboration with Terrapin Trader, ensuring devices were properly processed and documented.
- Maintained inventory records and tracked hardware assets using ServiceNow Hardware Asset Management to support accurate equipment management and operational efficiency.
- Collaborated with full-time staff and Service Desk Administrators to organize physical workspaces, support internal workflows, and maintain reliable technical support operations.

## PROJECTS

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**SBGP MWBE Information Process Mapping**, *Information Science Capstone* Spring 2026

- Analyzed and redesigned the South Baltimore Gateway Partnership's M/WBE compliance workflow to improve vendor experience, reduce administrative inefficiencies, and streamline internal review processes.
- Conducted stakeholder interviews with operations and program teams to identify workflow bottlenecks, communication gaps, and compliance tracking challenges across the vendor submission lifecycle.
- Developed current and optimized workflow diagrams, vendor-facing compliance guides, submission checklists, and redesigned budget templates to standardize onboarding and reduce revision cycles.
- Created a technical tool integration plan evaluating workflow automation, centralized tracking, and CRM integration opportunities using platforms such as Asana, Microsoft Power Platform, and Salesforce CRM.

**VitaMetrics: Personal Health Dashboard**, *Cloud Computing* Spring 2026

- Developed a full-stack cloud-based health tracking dashboard that enables users to monitor sleep, exercise, nutrition, mood, and wellness metrics.
- Implemented dynamic dashboard filtering, goal tracking, nutrition analytics, and file management features using React, TypeScript, and Express.
- Designed backend APIs with Node.js and Express, integrating Azure Cosmos DB and Azure Blob Storage to support scalable data storage and file upload workflows.
- Integrated Supabase Authentication for account management and configured GitHub Actions CI/CD pipelines to

automate Azure App Service deployment updates.

**Diabetes Risk Prediction & Fairness Analysis, *Applied Machine Learning & Model Evaluation***

Spring 2026

- Conducted exploratory and predictive analysis on 250,000+ rows of CDC BRFSS diabetes health indicator data to examine how lifestyle, cardiovascular, and demographic factors influence diabetes risk.
- Built and compared Decision Tree, SVM, KNN classification models in R, evaluating performance using ROC/AUC, sensitivity, specificity, and cross-validation metrics.
- Determined that the SVM model achieved the strongest predictive performance while assessing fairness across demographic groups to evaluate equitable healthcare risk prediction.

**PCOS Risk Prediction Model, *Applied Machine Learning & Model Evaluation***

Fall 2025

- Analyzed clinical and lifestyle data from 500+ patients to identify predictors of PCOS risk, define high-risk groups, and support early screening strategies.
- Applied K-means clustering using Scikit-learn in Python to uncover common patient risk profiles for targeted intervention.
- Built and evaluated a Random Forest classification model, assessing class imbalance and model performance. Interpreted feature importance and misclassifications to connect model behavior to real-world clinical patterns.

**Heart Disease Risk Cluster Analysis, *Data Science Techniques***

Fall 2025

- Conducted an exploratory analysis on 400,000+ rows of CDC BRFSS cardiovascular health data to examine how clinical indicators group into distinct heart disease risk profiles.
- Applied K-means clustering to visualize cluster separation and interpret underlying relationships between risk factors.
- Identified statistically distinct patient subpopulations and analyzed deviations in clinical indicators to uncover meaningful cardiovascular risk patterns.

**NHANES Health Data Analysis & Dashboard, *Data Visualization & Presentation***

Fall 2025

- Cleaned, transformed, and merged 20,000+ rows of NHANES survey data to assess the impact of socio-economic status, lifestyle choices, and demographic factors on health outcomes.
- Developed an R Shiny application with interactive filters and visualizations to communicate findings clearly to non-technical audiences.

**World Bank Life Expectancy Analysis, *Data Sources & Manipulation***

Fall 2025

- Performed exploratory and statistical analysis on World Bank global development indicators, examining relationships between life expectancy, healthcare expenditure, education spending, and income group.
- Applied linear regression and one-way ANOVA to validate trends observed in visualizations.

**DC Metro Ridership Analysis, *Info Challenge***

Spring 2025

- Cleaned, transformed, and validated 400,000+ rows of WMATA ridership data as part of an Agile team. Conducted exploratory data analysis to identify seasonal, station-level, and line-level usage patterns.
- Built a dynamic pricing script using NumPy and Scikit-learn to adjust advertising prices based on ridership trends.
- Visualized insights using Pandas and Matplotlib to support decision-making for transit planners and local businesses.

**Spotify Live Concert Business Strategy, *Smith Analytics Consortium's 6th Annual Datathon***

Spring 2025

- Crafted a data-backed proposal with a team for a 6-hour concert lineup by applying Scikit-learn k-means clustering on 2024's top-streamed songs to identify high-engagement genres and artist tiers.
- Built a Python-based dynamic pricing model using Capital One Arena seating data to estimate revenue with seat-section multipliers.
- Projected \$3.2M in ticket sales by integrating seat capacity and average prices, visualizing the trends with Matplotlib bar and pie charts.

- Proposed a summer launch with branding and marketing strategies tailored to student availability and seasonal demand.

#### **IT Job Postings Database, Database Design & Modeling**

Spring 2024

- Engineered a relational database to examine IT job market patterns using a 37,000+ record dataset, applying data modeling, normalization, and schema refinement to ensure efficiency and accuracy.
- Utilized forward and reverse engineering in MySQL Workbench to design, implement, and evaluate database structures. Queried the database to extract insights on salary distribution, skill demand, and company metrics.
- Ensured data integrity across 7 interconnected tables, collaborating in an Agile team to resolve design challenges and enhance performance.

#### **BestBefore: Expiration Tracking Web App, Dynamic Web Applications**

Spring 2025

- Deployed a full-stack web application on Vercel to help users track grocery expiration dates and receive recipe suggestions, reducing food waste and saving money.
- Integrated third-party APIs (TheMealDB, Recipe API) to dynamically fetch and display recipes based on user-inputted ingredients and expiration dates.
- Built interactive front-end interfaces with HTML, CSS, and JavaScript. Established back-end with custom API endpoints and Supabase for ingredient data storage.

#### **Screen Time vs. Academic Performance Analysis, Data Ethics**

Spring 2025

- Participated in a team research project analyzing survey data from 67 UMD students to examine relationships between phone screen time, study habits, and GPA. Ensured data anonymity and validity through structured consent, standardized formatting, and rigorous cleaning.
- Applied Spearman correlation, chi-squared tests, and Kruskal-Wallis hypothesis testing to assess associations between behavioral patterns and academic performance. Documented research and outcomes in a formal report.

#### **Cybersecurity Education Platform UX Research & UI Design, User-Centered Design**

Spring 2025

- Facilitated a research project on how non-technical users understand and manage digital privacy. Designed and distributed a survey across age groups to assess cybersecurity awareness, analyzing 29 responses through Power BI and qualitative coding.
- Identified key user personas and behavioral patterns through affinity diagrams. Prototyped a web-based learning platform in Figma and conducted usability testing to inform accessibility, content clarity, and personalized learning features.

#### **SafeTeal App & Website, Information Organization**

Spring 2024

- Collaborated in a team to design a location-based reporting platform in Figma that enables users to anonymously report and view incidents of inappropriate behavior.
- Applied information architecture, metadata organization, and iterative UX design principles to improve navigation, searchability, and user accessibility. Published an interactive prototype on GitHub.

#### **The Next Best NFL Team, Smith Analytics Consortium's 5th Annual Datathon**

Spring 2024

- Collaborated in a 7-member analytics competition team to evaluate expansion opportunities for a hypothetical NFL franchise using business intelligence and market analysis techniques.
- Used Tableau and Google Colab to analyze demographic, economic, and sports market data while developing revenue projections, promotional strategies, and presentation materials to support final recommendations.

#### **Personal Blog Website, Python Bootcamp**

Winter 2024

- Built a multi-page, responsive blog website using Flask, HTML, CSS, and SQLAlchemy, featuring secure authentication and CRUD functionality.
- Enabled registered users to create, edit, and delete blog posts. Integrated databases for storing post and user data. Ensured code reusability and modularity through RESTful route design and component-based templates.

#### **Portfolio Website, Personal Project**

Summer 2023

- Designed and developed a responsive personal portfolio website using HTML, CSS, SCSS, and JavaScript to showcase technical projects and design work.
- Implemented a mobile-first layout, scalable content structure, and reusable styling components to improve usability and long-term maintainability.

#### **Lemonaid: Health Tracking Web App, Bitcamp Hackathon**

Spring 2023

- Collaborated in a hackathon team to develop a wellness-tracking web application focused on user engagement and healthy habit monitoring.
- Integrated an AI-powered chat feature via API and designed intuitive user flows using HTML, CSS, and JavaScript to improve accessibility and interaction.

#### **Programming and Statistical Computing Projects, Object-Oriented Programming & Applied Statistics**

Fall 2023

- Developed multiple Java applications applying object-oriented design, data structures, and file handling, including a Media Rental Manager and Video Playlist Generator, improving data processing efficiency and user experience.
- Applied R for data simulation, statistical testing, and distribution modeling, calculating estimators, confidence intervals, and normality measures with built-in datasets. Combined statistical analysis and programming to manipulate structured data and identify trends.

### **EXTRACURRICULARS**

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#### **VP of Recruitment, Alpha Sigma Kappa, Women in Technical Studies**

Fall 2025

- Led chapter recruitment initiatives by planning and coordinating information sessions, tabling events, and recruitment activities designed to foster an inclusive and welcoming environment for prospective members.
- Served as the primary point of contact for recruitment inquiries, communicating with students and guiding them through the recruitment process.
- Collaborated with executive board members to organize recruitment events, manage outreach efforts, and support engagement strategies across campus.
- Oversaw the development and approval of recruitment-related promotional materials, including digital graphics, flyers, and apparel, to strengthen chapter visibility and branding.

#### **Design Team Organizer, Bitcamp Hackathon**

Spring 2025

- Collaborated with Bitcamp's design and development teams to improve the usability, accessibility, and visual consistency of the hackathon website and digital platforms.
- Contributed to UI/UX design, graphic design, social media content, branding assets, and constructive design critiques in preparation for Maryland's annual hackathon event.
- Supported event logistics during the hackathon by assisting participants, managing shifts and outreach activities, evaluating projects, and helping create a welcoming and organized experience for hundreds of attendees.

#### **Social Media & Marketing Chair, Alpha Sigma Kappa, Women in Technical Studies**

Fall 2024

- Managed the sorority's digital presence across Instagram, TikTok, and the chapter website by developing content strategies, coordinating posting schedules, and maintaining up-to-date event and recruitment information.
- Designed graphics, videos, photography, and written content to promote chapter events, philanthropy initiatives, member achievements, and recruitment efforts in a cohesive and engaging way.
- Collaborated with recruitment, philanthropy, and event planning teams to support outreach campaigns and strengthen member engagement through social media marketing.

### **SKILLS**

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**Languages:** C++, HTML, CSS, JavaScript, Java, Python, SQL, R

**Softwares:** Microsoft Office, Eclipse, PyCharm, Matlab, MySQL Workbench, R Studio, Visual Studio Code, GitHub, Tableau, PowerBI, Jupyter Notebook, Azure, AWS